

Barrier Options

(1. First Passage Times)

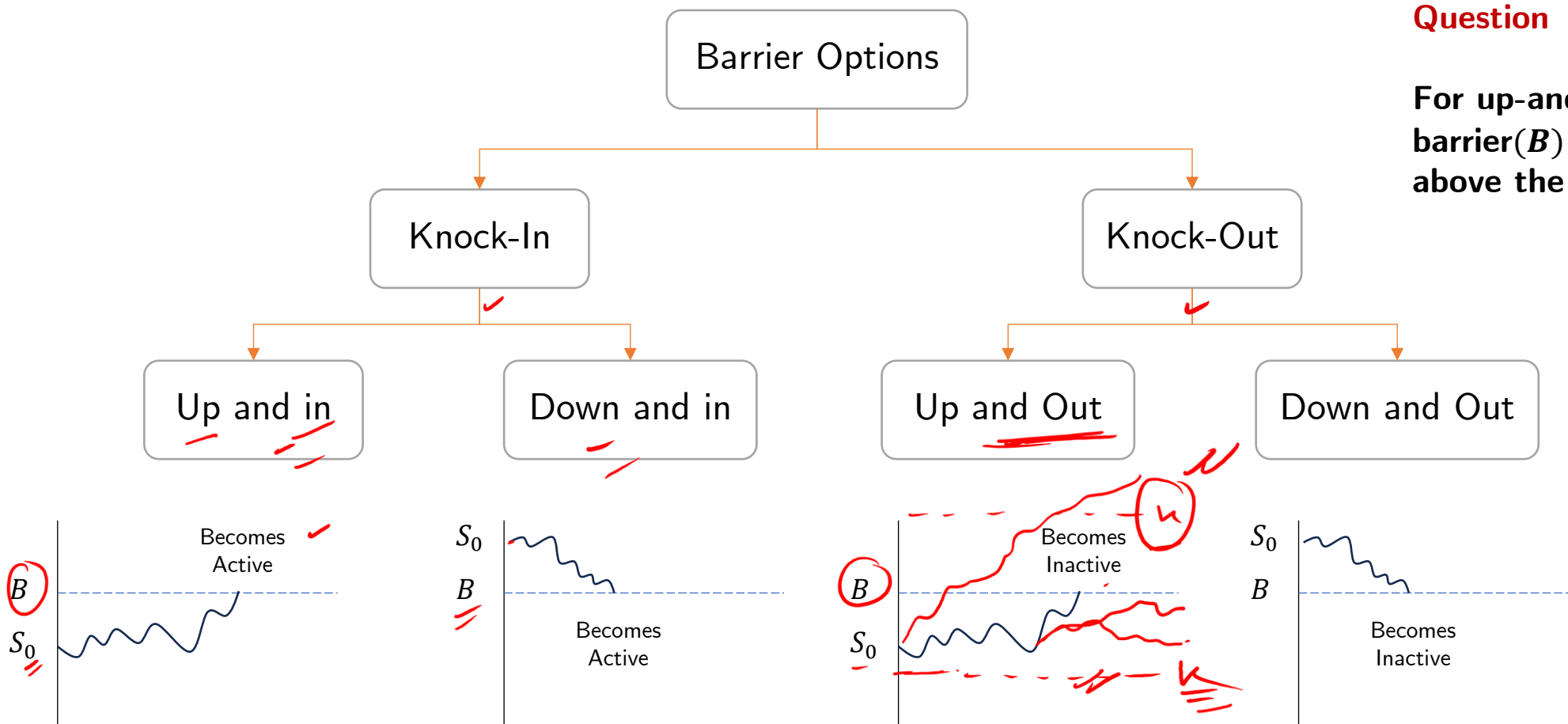
Barrier Options

Barrier options come in the following flavors

$$\text{Payoff} = (S_T - K)^+$$

Question

For up-and-out Call, the barrier (B) should be kept above the strike (K). (T/F) ?



MGF and PGF

We spoke about **MGF (Moment Generating Function)** earlier. There is a related power series applicable for discrete random variables, called as PGF (Probability Generating Functions)

(MGF) $M_X(t) = \mathbb{E}[e^{tX}] = 1 + t\mathbb{E}[X] + \frac{1}{2}t^2\mathbb{E}[X^2] + \dots$ (1)

(Handwritten notes: $\frac{1}{2} \cdot 2t = t$)

$\mathbb{E}[X] = \left. \frac{dM}{dt} \right|_{t=0}$

$\mathbb{E}[X^2] = \left. \frac{d^2M}{dt^2} \right|_{t=0}$

$X = 0, 1, 2, 3, \dots$

(PGF) $G_X(z) = \mathbb{E}[z^X] = \sum_{x=0}^{\infty} p(x)z^x = \underbrace{z^0}_{1}P(X=0) + \underbrace{z^1}P(X=1) + \underbrace{z^2}P(X=2) + \dots$

(Handwritten notes: $= P(X=0) + zP(X=1) + z^2P(X=2) + \dots$)

(Handwritten note: z^0)

The coefficient of z^n , denotes the probability of X taking a value n, i.e., $P(X = n)$

Properties of PGF

Here are some important properties of PGF

$$G_X(z) = \mathbb{E}[z^X] = \underline{P(X=0)} + \underline{zP(X=1)} + \underline{z^2P(X=2)} + \dots$$

1) The probability mass function of X is recovered by taking derivative of G , $P(X=k) = \frac{G^{(k)}(0)}{k!}$

2) Total probability = 1, $G_X(1) = 1$

3) Expectation can be recovered as $\mathbb{E}[X] = G'(1)$

4) Sum of **independent randoms** $G_{X_1+X_2}(z) = G_{X_1}(z) \cdot G_{X_2}(z)$

5) PGF uniquely determines the probability distribution of X

$$G''(z) = 2 \cdot P(X=2) + \dots$$

$$G^{(k)}(z) = \frac{k! \cdot P(X=k)}{k!} = P(X=k)$$

$$G_X(0) = P(X=0)$$

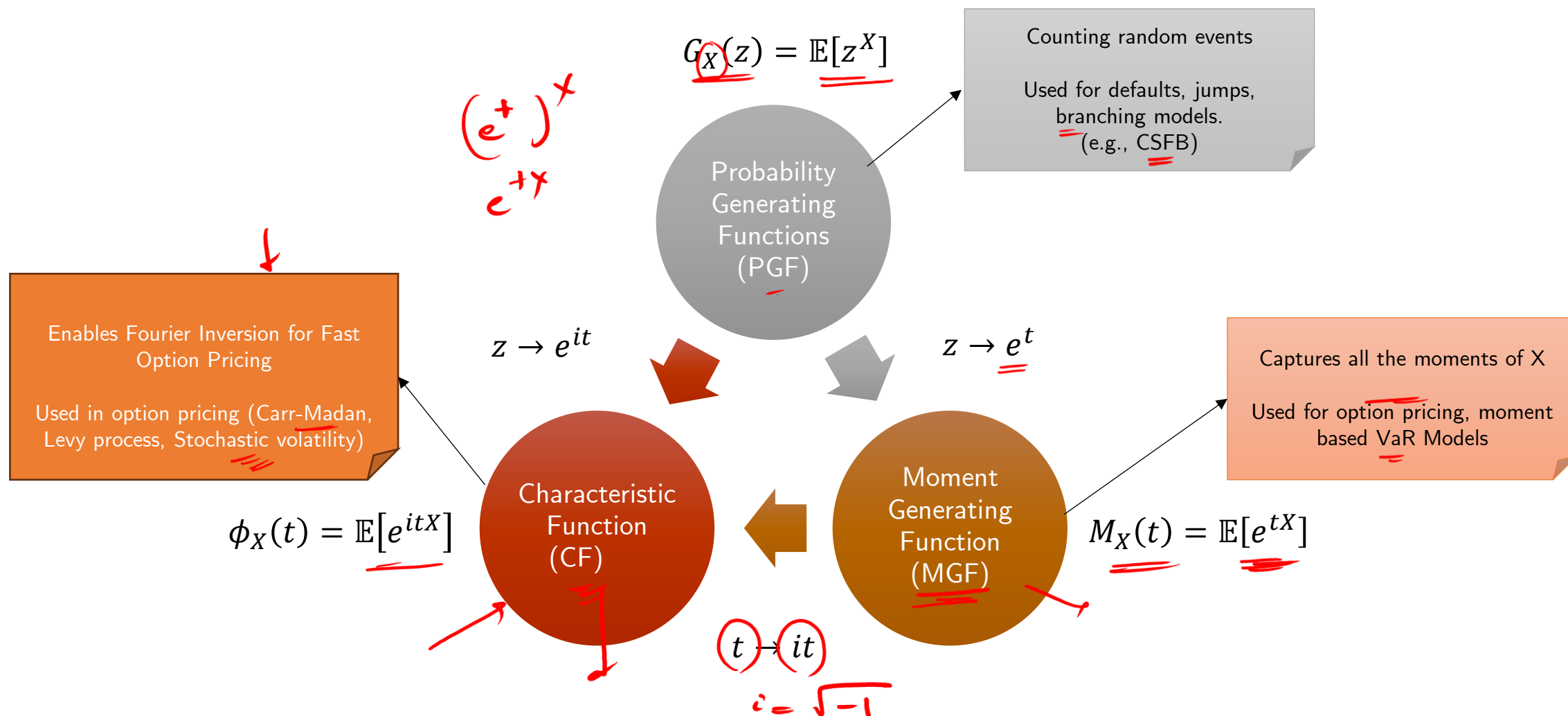
$$G_X(1) = 1$$

$$\mathbb{E}[X] = G'(1) = P(X=1) + 2zP(X=2) + 3z^2P(X=3) + \dots$$

$$\mathbb{E}(X) = 1 \cdot P(X=1) + 2 \cdot P(X=2) + 3 \cdot P(X=3) + \dots$$

$$\mathbb{E}[z^{X_1+X_2}] = \mathbb{E}(z^{X_1} \cdot z^{X_2}) = \mathbb{E}(z^{X_1}) \cdot \mathbb{E}(z^{X_2})$$

MGF, PGF, CF



Exercise

1) Find the PGF of

a) Bernoulli Random and

b) Binomial Random

$$1) X \sim \text{Bernoulli}(p) \quad G_X(z) = \mathbb{E}[z^X] = z^1 \cdot p + z^0 \cdot (1-p) = \underline{(pz + 1-p)}$$

$$X \begin{cases} \rightarrow 1, p \\ \rightarrow 0, 1-p \end{cases}$$

$$G_Y(z) = G_{X_1+X_2+\dots+X_n}(z) = (pz + 1-p)^n$$

$$Y \sim \text{Binomial}(n, p) \quad \text{i.i.d.} \\ Y = X_1 + X_2 + \dots + X_n, \quad X_i \sim \text{Bernoulli}(p)$$

2) Find the PGF of Poisson distribution. Derive the Expectation from PGF.

$$X \sim \text{Poi}(\lambda)$$

$$\underline{G_X(z)} = \mathbb{E}[z^X] = \underline{z^0} P(X=0) + z^1 P(X=1) + z^2 P(X=2) + \dots$$

$$= \frac{e^{-\lambda} \lambda^0}{0!} + z \cdot \frac{e^{-\lambda} \lambda^1}{1!} + z^2 \cdot \frac{e^{-\lambda} \lambda^2}{2!} + \dots$$

$$G = e^{\lambda(z-1)}$$

$$G' = e^{\lambda(z-1)} \cdot \lambda$$

$$G'(1) = \lambda$$

$$= e^{-\lambda} \left(1 + \frac{(z\lambda)}{1!} + \frac{1}{2!} (z\lambda)^2 + \dots \right) \\ = e^{-\lambda} \cdot e^{z\lambda} = e^{\lambda(z-1)}$$

$$\begin{pmatrix} \mathbb{E}(X^2) \\ \mathbb{E}(X) \end{pmatrix}$$

$$V(X) = \mathbb{E}(X^2) - \mathbb{E}(X)$$

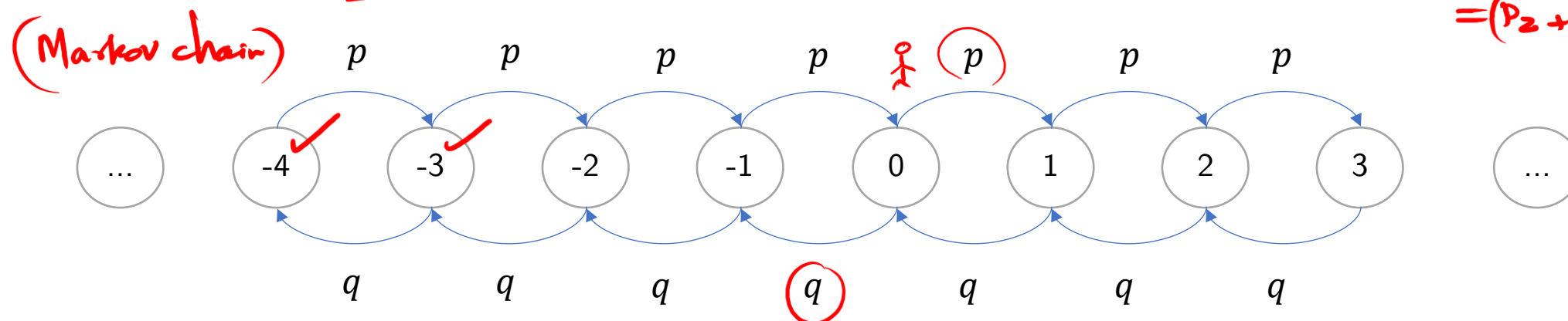
Random Walk

Recall that a random walk is a series of i.i.d steps where the increments can be either +1 or -1.

(i.i.d increments) $X_i = \begin{cases} 1, & \text{prob} = p \\ -1, & \text{prob} = q \end{cases}$ $p + q = 1$

Walker $S_n = X_1 + X_2 + \dots + X_n$, $S_0 = 0$

$$G_{X_i}(z) = \mathbb{E}(z^{X_i}) \\ = p \cdot z^1 + q \cdot z^{-1} \\ = (pz + qz^{-1})$$



Show that the PGF of the walk is $G_{S_n}(z) = (pz + qz^{-1})^n$

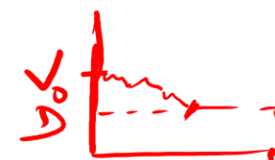
$$G_{S_n}(z) = G_{X_1 + X_2 + \dots + X_n}(z) = (pz + qz^{-1})^n$$

First Passage Time (I)

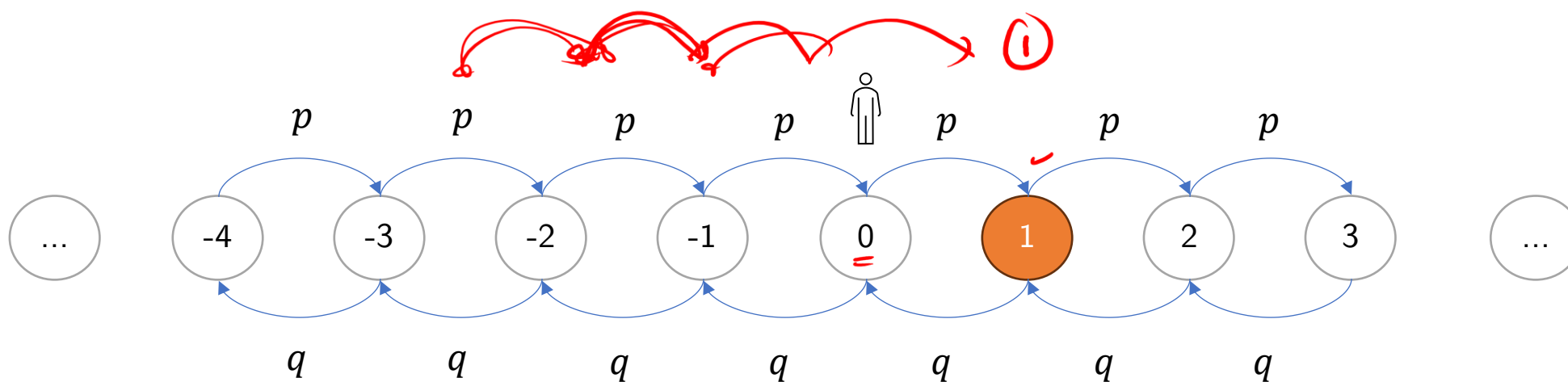
Given the Simple Random Walk, we want to find out the following

The distribution of first time the walker reaches (+1). This is called as 'hitting time' or 'first passage time'

S_n



Define the stopping time as $\tau = \min \{S_k = 1 | S_0 = 0\}$. Let's derive the PGF of τ



First Passage Time (II)

The PGF of τ is given as $G_\tau(z) = \mathbb{E}[z^\tau]$

Let's do one-step conditioning $G_\tau(z) = \mathbb{E}[\mathbb{E}(z^\tau | \text{one step})]$

$$= p[\mathbb{E}(z^\tau | \text{right})] + q[\mathbb{E}(z^\tau | \text{left})]$$

$$= pz^1 + q[\mathbb{E}[z^{1+\tau'+\tau''}]] \rightarrow$$

$$= pz + qzG_\tau(z)^2$$

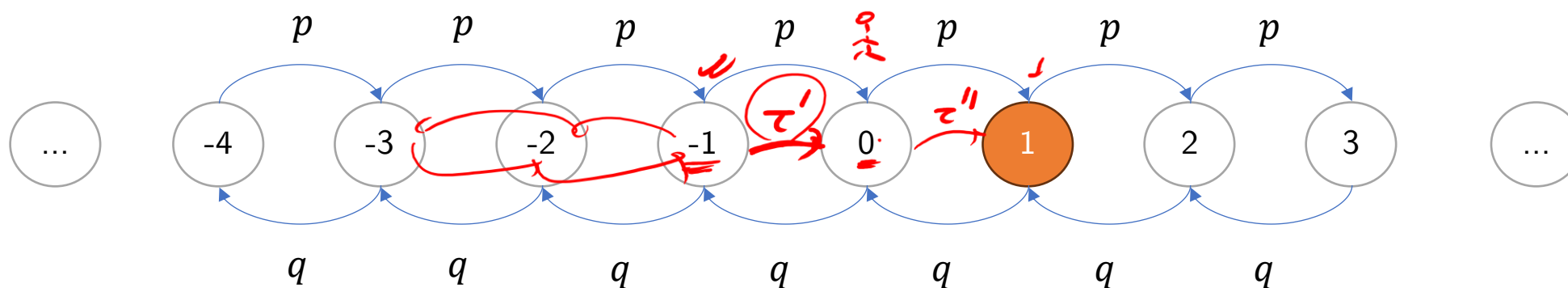
τ' and τ'' are the copies of τ indicating
Hitting times from -1 to 0 and 0 to 1

Characteristic equation :

$$qz\alpha^2 - \alpha + pz = 0$$

where $\alpha = G_\tau(z)$

$$\begin{aligned} \mathbb{E}((z^1)z^{\tau'}z^{\tau''}) &= z\mathbb{E}(z^{\tau'}) \cdot \mathbb{E}(z^{\tau''}) \\ &= z \cdot \mathbb{E}(z^\tau) \cdot \mathbb{E}(z^\tau) \\ &= z \cdot (\mathbb{E}(z^\tau))^2 \end{aligned}$$



First Passage Time (III)

Characteristic equation : $qz\alpha^2 - \alpha + pz = 0$ ✓

Solving $\alpha = \frac{1(\pm)\sqrt{1-4pqz^2}}{2qz} = G_\tau(z)$
 $z \rightarrow 0$

Since α is a PGF, we need to ensure it does not blow up for small values of z . We must take

$$G_\tau(z) = \frac{1 - \sqrt{1 - 4pqz^2}}{2qz}$$
 ✓

$p=q=1/2$

For simple symmetric random walk, where $p = q = \frac{1}{2}$

$\mathbb{E}(\tau)$ $\mathbb{E}(z^\tau)$

$$G_\tau(z) = \frac{1 - \sqrt{1 - z^2}}{z}$$
 ✓

Probability of hitting

Are we guaranteed to hit +1 in finite time ? In other words, does the first passage occur at all ? We are looking for probability $P(\tau < \infty)$. Note that $P(\tau < \infty) = G_\tau(1)$. Using $p + q = 1$

$$\tau = \infty$$

$$\tau = 0$$

$$\tau = 1$$

$$\tau = 2$$

$$\vdots$$

$$G_\tau(1) = \frac{1 - \sqrt{1 - 4pq}}{2q}$$

$$G_\tau(z) = \frac{1 - |p - q|}{2q}$$

$$\begin{aligned} & \frac{\sqrt{1 - 4pq}}{(p+q)^2 - 4pq} \\ &= \sqrt{(p-q)^2} \\ &= |p-q| \end{aligned}$$

$$\tau = \infty$$

$$1 - (p/q)$$

$$G_\tau(1) = \frac{p}{q} < 1$$

$$p > q$$

$$G_\tau(1) = 1$$

$$\frac{1 - p + q}{2q}$$

$$\frac{1 - q + p}{2q}$$

If $p > q$ (in other words, there is a +ve drift in the process), then the hitting +1 is guaranteed. However, if $p < q$ (negative drift), then there is non-zero probability $\left(1 - \frac{p}{q}\right)$ that $\tau = \infty$ (first passage never occurs)

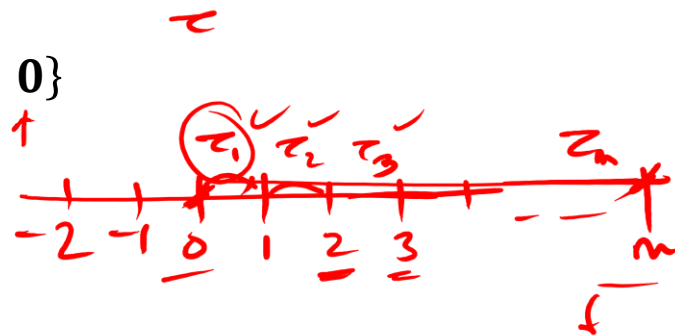


General First Passage Times

What is the distribution of hitting times for a level $= +m$?

Define $\tau_m = \min \{S_k = m | S_0 = 0\}$

The PGF is given as



$$G_{\tau_m}(\mathbf{z}) = \left(\frac{1 - \sqrt{1 - 4pqz^2}}{2qz} \right)^m$$

Expectation : $\mathbb{E}(\tau_m)$

What is the expected value of the hitting time for the barrier = m ? Since, we have the PGF, we can get the expected hitting time by taking

$$\mathbb{E}(\tau) = \tau = \boxed{0} \times (1 - p/q)$$

$$\mathbb{E}[\tau_m] = G'_{\tau_m}(1)$$

$$z=1$$

$$z^n \rightarrow P(\tau=n)$$

$$G_{\tau_m}(z) = \left(\frac{1 - \sqrt{1 - 4pqz^2}}{2qz} \right)^m$$

$$p \leq q$$

$$p > q$$

$$\mathbb{E}[\tau_m] = \infty$$

$$\mathbb{E}[\tau_m] = G'_{\tau_m}(1) = \frac{m}{p - q}$$

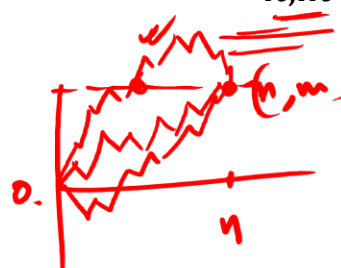
$$p \gg q \quad p \rightarrow q$$

$$p=q$$

Probability : $\mathbb{P}(S_n = m)$

What is the probability the walker will be at level (m) after (n) steps ? Let's answer a simpler question first. Is it possible to reach level m after n steps ? If yes, what is the probability of reaching it ?

Define $N_{n,m}$ = (Total paths that reach level m after n steps). Then



Handwritten notes and equations:

$$N_{n,m} = \binom{n}{\frac{n+m}{2}} = \binom{n}{\frac{n-m}{2}}$$

$$\mathbb{P}(S_n = m) = P_{n,m} = \binom{n}{\frac{n+m}{2}} p^{\frac{n+m}{2}} q^{\frac{n-m}{2}}$$

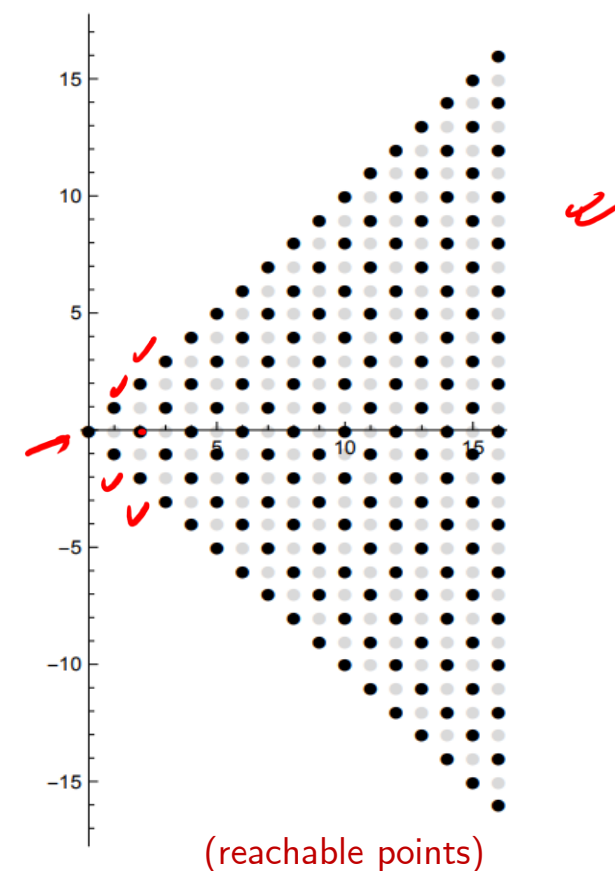
Handwritten derivation:

$$x(1) + (n-x)(-1) = m$$

$$\Rightarrow x - n + x = m$$

$$\Rightarrow x = \frac{n+m}{2}$$

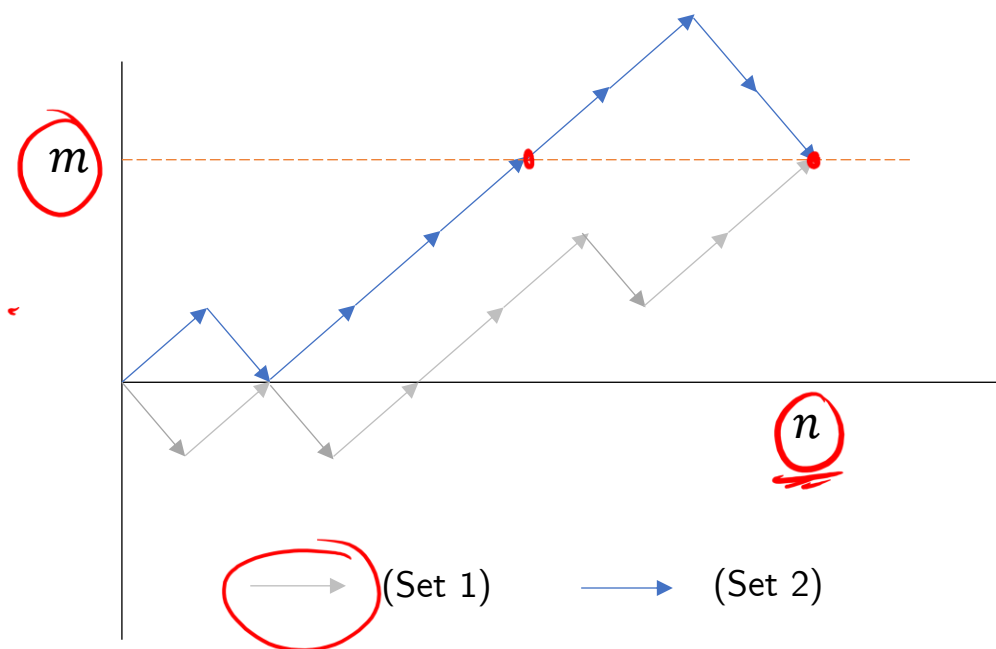
Note that the reachability condition is given as $n \equiv m \pmod{2}$



Probability : $\mathbb{P}(\tau_m = n)$

All the paths that lead from origin to level = m , in n steps i.e., $N_{n,m}$ consist of two sets of paths

- 1) **(Set 1)** Paths which visit level = m for the first time at step = n (This is our hitting time)
- 2) **(Set 2)** Paths which visits level = m before step = n and then return to level = m at step = n



The probability of first passage (level m) taking a value n is a fraction of the probability of being at level m at time n . The formula is given by **Ballot principle**.

$$\mathbb{P}(\tau_m = n) = \left(\frac{m}{n}\right) \mathbb{P}(S_n = m)$$



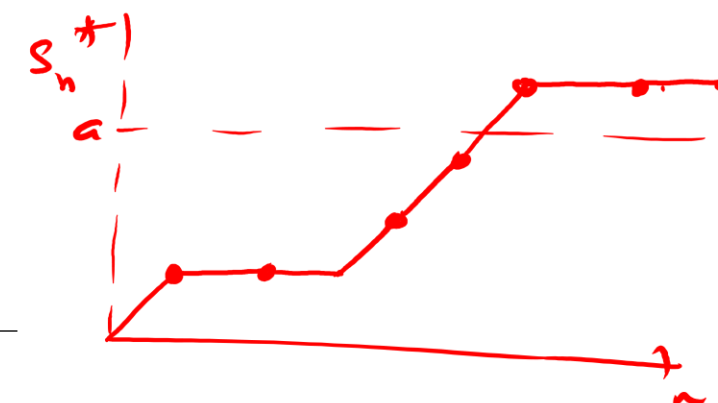
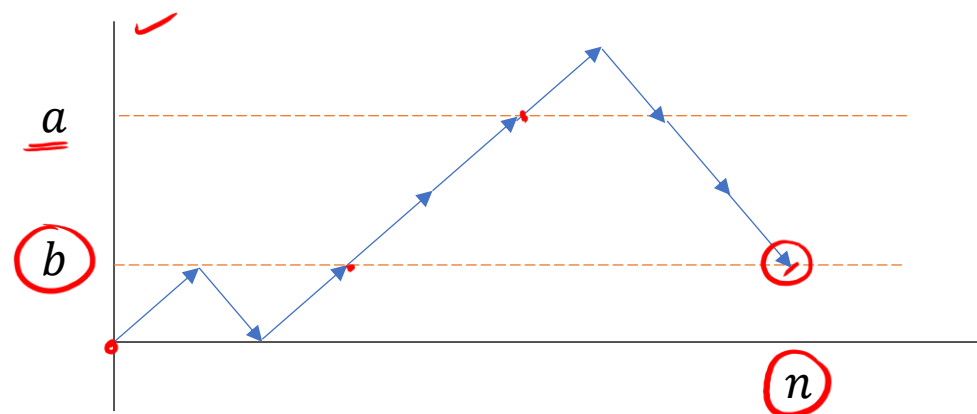
Reflection Principle (I)

Consider a simple symmetric random walk. What is the probability that the walker crosses the barrier a at some interim step but ends at $b \leq a$ after n steps? In other words, define a running maximum

$$S_n^* = \max_{1 \leq j \leq n} S_j$$

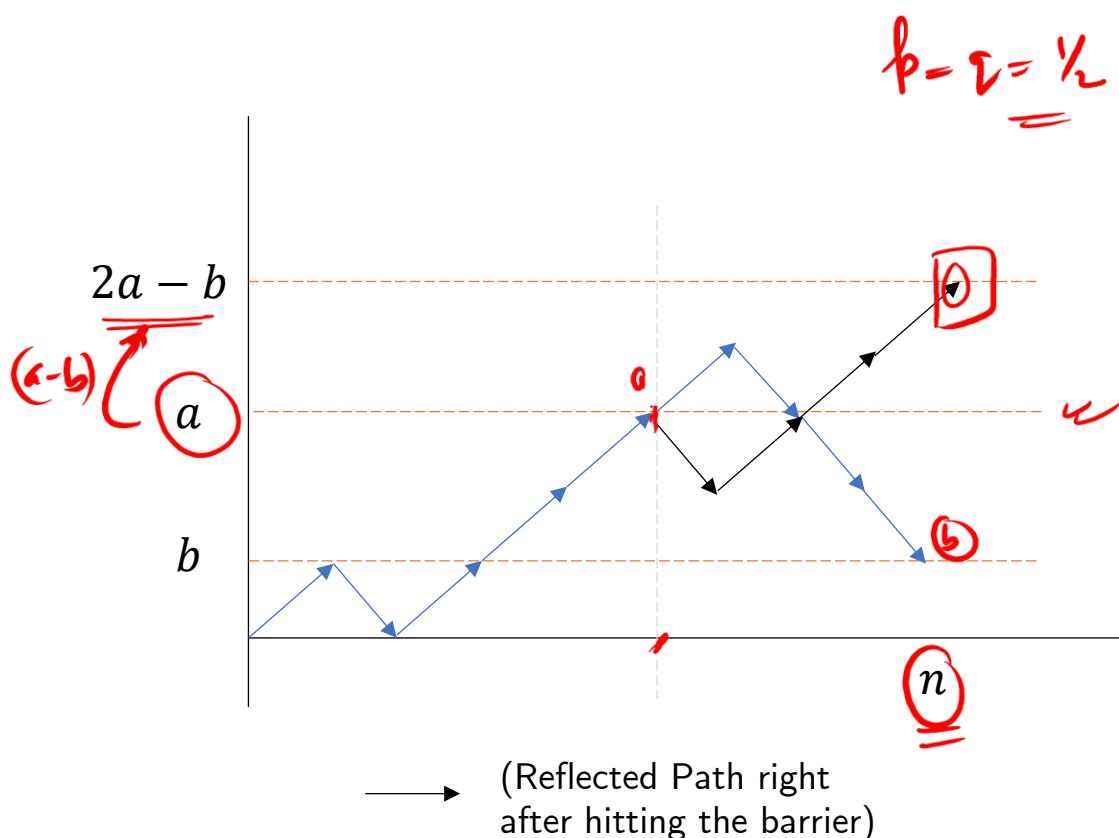
Find the probability $\mathbb{P}(S_n^* \geq a, S_n = b)$

③



Reflection Principle (II)

Consider a simple symmetric random walk. What is the probability that the walker crosses the barrier a at some interim step but ends at $b \leq a$ after n steps? In other words, define a running maximum



There is a bijection between the original paths that satisfy the above question and the reflected paths right after hitting the barrier a all of which must end at $2a - b$

$$\mathbb{P}(S_n^* \geq a, S_n = b) = \mathbb{P}(S_n = 2a - b)$$

$P(S_n = m)$